ENGR 1451/2451: Exploratory Data Science

Descriptive Statistics Review

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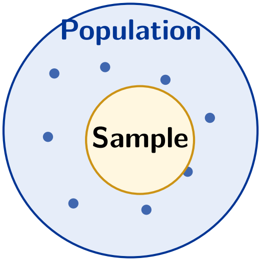
Population, Sample, and Inference



- **Descriptive statistics** summarize observed data.
- **Inferential statistics** use a sample to learn about a population.
- **Probability** describes sample-to-sample variation.
- Inference is credible only when the sample and measurements are appropriate for the question.

Notation

Symbol	Meaning
n and N	sample size and finite-population size
$x_1, x_2, \ldots, x_n$	individual observations
$\bar{x}$ and μ	sample mean and population mean
s^2 and σ^2	sample variance and population variance
s and σ	sample standard deviation and population standard deviation
Q_1, Q_2, Q_3	first quartile, median, and third quartile

Measures of Center

Sample mean

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

- Arithmetic balance point
- Uses every observation
- Sensitive to extreme values

Median

Middle value after ordering; divides the distribution into two equal halves.

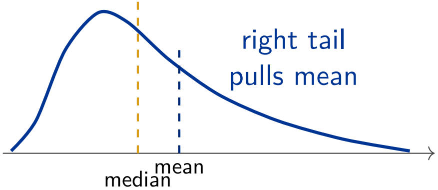
Mode

Most frequently occurring value; especially useful for categorical data.

Skewness Determines Mean vs. Median

Shape	Relationship
Symmetric	mean $\approx$ median
Right-skewed	mean $>$ median
Left-skewed	mean $<$ median

The tail pulls the mean farther than it moves the median.



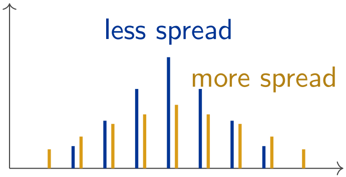
Choose Summaries to Match the Distribution

Distribution	Center	Spread
Roughly symmetric, no strong outliers	Mean	Standard deviation
Skewed or containing strong outliers	Median	Interquartile range

Why Variability Matters

- Similar centers can coexist with very different spreads.
- Variability describes how much observations differ from one another.
- The range is simple but uses only two observations:

$$\text{Range} = \max(x) - \min(x)$$



Variance and Standard Deviation

Describe a complete list or population

$$\sigma^2 = \frac{\sum_{i=1}^{N}(x_i - \mu)^2}{N}, \qquad \sigma = \sqrt{\sigma^2}$$

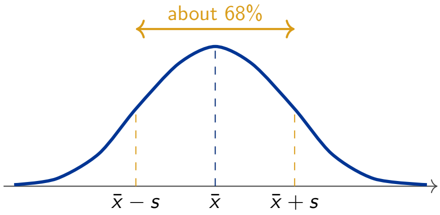
Estimate population spread from a sample

$$s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}, \qquad s = \sqrt{s^2}$$

- Squaring prevents positive and negative deviations from canceling.
- Variance is in squared units; standard deviation returns to the original units.
- State the convention used for small datasets.

Interpreting Standard Deviation

- SD is a typical horizontal distance from the mean.
- Estimate it from horizontal spread, not peak height.
- For bell-shaped data, about 68% lies within 1 SD and about 95% within 2 SDs.
- The 68–95 rule is not universal.



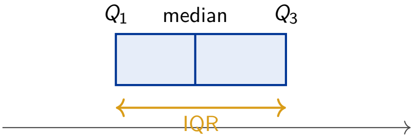
Shifts, Rescaling, and Reflection

Transformation	**Effect on center**	**Effect on spread**
Add c to every value	mean and median increase by c	SD and IQR do not change
Multiply every value by c	center is multiplied by c	SD and IQR are multiplied by $\lvert c \rvert$
Reflect, $y = a - x$	mean shifts; skew direction reverses	SD and IQR are unchanged

Quartiles and Interquartile Range

- Q_1: 25th percentile
- Q_2: median, or 50th percentile
- Q_3: 75th percentile
- The middle half of the observations lies between Q_1 and Q_3.

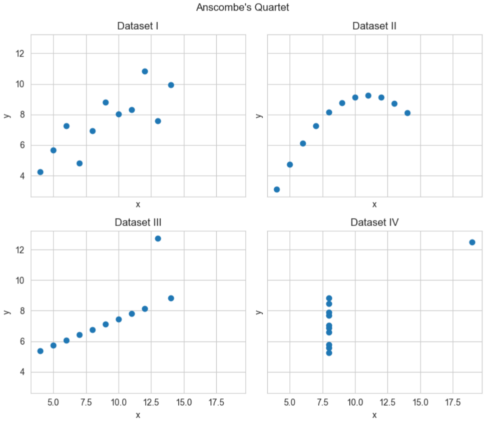
$$\mathrm{IQR} = Q_3 - Q_1$$



Data Visualization

- Scatter plot
- Histogram
- Boxplot

Same Summaries, Different Data



Anscombe's lesson

Datasets can have nearly identical means, SDs, and correlations but very different relationships.

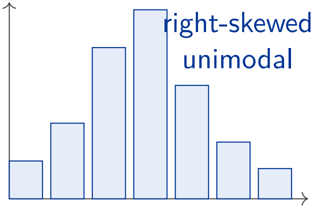
Plot first; interpret summaries in the context of the plot.

Histograms: Bins, Area, and Shape

- A histogram groups one quantitative variable into bins.
- Bar area represents count or proportion in the interval.
- With unequal widths, use

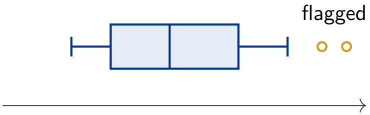
$$\text{density} = \frac{\text{proportion in bin}}{\text{bin width}}.$$

- Describe modality, skewness, clusters, gaps, and outliers.
- Bin choice can reveal or obscure structure.



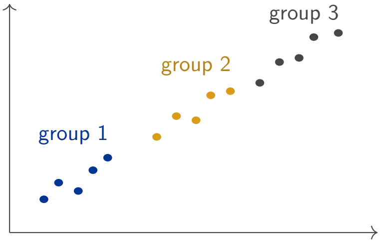
Boxplots and Outliers

- Box: Q_1 to Q_3; line: median.
- Whiskers extend to the most extreme non-outliers.
- Flag values below $Q_1 - 1.5\,\mathrm{IQR}$ or above $Q_3 + 1.5\,\mathrm{IQR}$.

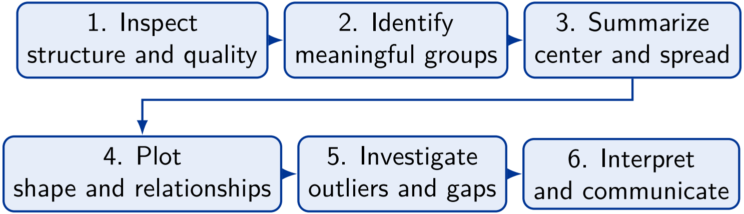


Scatter Plots and Group Structure

- Each point represents paired values of two quantitative variables.
- Look for direction, form, strength, clusters, gaps, and outliers.
- Color or symbols can reveal categorical groups.
- A pooled trend or mean may be unrepresentative when groups differ.



A Compact Exploratory Workflow



Reproducibility

The notebooks should run from the first cell to the last and preserve the code, output, and interpretation used to reach each conclusion.

Summary

- Begin with observations, variables, types, groups, and data quality.
- Use mean and SD for roughly symmetric data without strong outliers; use median and IQR for skewed or outlier-prone data.
- Variance and SD use deviations from the mean; distinguish the N and $n - 1$ conventions.
- Histograms reveal shape, boxplots compare groups and flag outliers, and scatter plots reveal relationships.
- Numerical summaries and plots are complements, not substitutes.

Questions?